

VICTORIA PARK ACADEMY SHENZHEN

#

IGCSE COMPUTER SCIENCE (0478) — YEAR 10

SEMESTER 2

#

WEEKS 1–9: INTENSIVE REVISION & MOCK

SERIES

#

Academic Year 2025–2026 | Final Semester

Before IGCSE Examinations

#

Course: CIE IGCSE Computer Science (0478) | Class: Year 10 | Semester: 2

Lesson Duration: 40 minutes | Student Profile: EAL learners, mixed ability

Total Weeks: 9 | Total Lessons: 18 lessons + 2 mock exam sessions

GRADE BOUNDARIES: A*:68-75 A:60-67 B:52-59 C:44-51 D:36-43 E:28-35 F:20-27 G:<20

TIME MANAGEMENT: Paper 1 & 2: 1h45m for 75 marks = ~1.4 min/mark.

2-mark Q: 2.8min | 4-mark Q: 5.6min | 6-mark Q: 8.4min | 8-mark Q: 11.2min

15-mark scenario: ~30 min | Reserve 5 min for checking.

CONTENTS:

UNIT 1 (Weeks 1-4): Paper 1 Intensive Revision

W1: Data Representation & Compression

W2: Data Transmission, Hardware & Software

W3: Internet, Security & Emerging Tech

W4: Paper 1 Timed Practice

UNIT 2 (Weeks 5-8): Paper 2 Intensive Revision

W5: Algorithms & Problem Solving

W6: Python Programming

W7: Databases, Boolean Logic & Validation

W8: Paper 2 Timed Practice + 15-Mark Scenario

UNIT 3 (Week 9): Mock Series 1

W9: Mock Exam 2 — Paper 1 (Full 75-mark paper + mark scheme)

CROSS-REFERENCE: Syllabus Topics 1-10

- T1: Types and components of computer systems
- T2: Input and output devices
- T3: Storage devices and media
- T4: Networks and the effects of using them
- T5: The effects of using ICT / Emerging technologies
- T6: ICT applications / Automated systems / Robotics / AI
- T7: Algorithm design (pseudocode, flowcharts, decomposition, abstraction)
- T8: Programming (Python, constructs, data structures, file handling)
- T9: Databases (SQL, normalisation, keys, ERDs)
- T10: Boolean logic (gates, truth tables, logic circuits, validation)

UNIT 1: PAPER 1 INTENSIVE REVISION



WEEK 1: DATA REPRESENTATION & COMPRESSION

Syllabus Ref: T1.1-1.7 | 2 lessons, 2 worksheets, answer keys

--- LESSON 1.1 — BINARY, HEXADECIMAL & FILE SIZE CALCULATIONS -----

Topic: Number Systems & File Sizes | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/1 1.1-1.4

Learning Objective: Convert denary<->binary (up to 16-bit), binary addition with overflow detection, hex conversions, calculate image/sound file sizes.

Key Vocabulary:

- binary (二进制 / erjinzhi) - denary (十进制 / shijinzhi)
- hexadecimal (十六进制 / shiliu jinzhi) - overflow (溢出 / yichu)
- bit depth (位深度 / wei shendu) - sampling rate (采样率 / caiyang lv)
- resolution (分辨率 / fenbian lv) - pixel (像素 / xiangsu)

Lesson Structure (40 min):

0-3 | DO NOW: 5 conversion questions on board. Students work silently.

3-12 | DIRECT INSTRUCTION: Division-by-2 method; place values (128..1);

nibble grouping for hex; 16-bit conversions; overflow detection

(carry into sign bit != carry out); image file size (WxHxD/8/1024);

sound file size (Rate x Depth x Duration x Channels)

12-25 | GUIDED PRACTICE: Worksheet 1A in pairs. 8-bit -> 16-bit progression.

Common errors: forgetting to pad with leading zeros, confusing colour depth with resolution.

25-35 | INDEPENDENT PRACTICE: 10 rapid-fire conversions, calculator-free.

35-40 | PLENARY: Quick-fire oral quiz. HW: Complete Worksheet 1A.

EAL Support: Bilingual keyword cards, visual place-value table (laminated A4),
buddy pairing, Chinese notation allowed in working.

Differentiation:

Emerging: 8-bit only, place-value table, teacher-checks-each-answer

Developing: 16-bit, binary addition, image size with formula, some calc use

Secure: 16-bit no aids, all file sizes, explain overflow, create peer questions

Resources: Slides, laminated place-value tables, Worksheet 1A.

--- LESSON 1.2 — COMPRESSION, TEXT & SOUND REPRESENTATION -----

Topic: Compression & Data Types | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/1 1.5-1.7

Learning Objective: Explain lossy vs lossless, apply RLE, explain Unicode
(relevance to Chinese), describe sound digitisation.

Key Vocabulary:

- lossy compression (有损压缩 / yousun yasuo)
- lossless compression (无损压缩 / wusun yasuo)
- RLE (行程编码 / xingcheng bianma) - ASCII - Unicode (统一码 / Tongyima)
- sampling (采样 / caiyang) - bit rate (比特率 / bite lv) - metadata

Lesson Structure (40 min):

0-3 | DO NOW: "A 64-bit binary image - how to make the file smaller?"

3-12 | DIRECT INSTRUCTION:

1. Lossy vs Lossless: Lossy removes data permanently (JPEG,MP3,MP4).
Lossless preserves all data (PNG,FLAC,ZIP). Trade-off: quality vs size.
2. RLE: AAAAABBCCCCDDDE -> A5 B2 C4 D5 E1. Effective for simple
graphics with large same-colour areas; NOT effective for photographs.
Check: if compressed > original, RLE not suitable.
3. ASCII=7-bit=128 chars (English only). Unicode=up to 32-bit=149,000+
chars. CHINESE CONNECTION: 50,000+ Chinese characters; ASCII cannot
represent ANY Chinese character. UTF-8: variable length - English=1
byte, Chinese=3 bytes.
4. Sound digitisation: Analog->sample at intervals->quantise to binary.
Nyquist Theorem: rate $\geq 2 \times$ highest frequency. CD: 44.1kHz, 16-bit,
stereo = 1411 kbps.

12-25 | GUIDED PRACTICE: RLE Compression Race (10x10 colour grids).

Unicode exploration: type name in English vs Chinese, compare file sizes.

25-35 | INDEPENDENT PRACTICE: Worksheet 1B - past paper style questions.

35-40 | PLENARY: Exit ticket (3 questions). HW: Complete Worksheet 1B.

EAL Support: Bilingual glossary, visual JPEG quality comparison, relate

Unicode to students' own language, RLE uses colours (accessible to all EAL).

Differentiation:

Emerging: Definitions/examples, matching, guided RLE, fill-in-blank

Developing: RLE encode/decode independently, explain trade-offs, calculate
sound files, compare ASCII/Unicode with examples

Secure: Evaluate RLE effectiveness, analyse ratios, explain UTF-8 variable
length, discuss quality vs size numerically

Resources: Slides, coloured grid templates, Worksheet 1B.

--- WORKSHEET 1A — BINARY, HEXADECIMAL & FILE SIZE CALCULATIONS -----

Total: 35 marks | Name: _____ Class: _____ Date: _____

SECTION A: BINARY CONVERSIONS (10 marks)

1. Convert to 8-bit binary: a) 45 b) 128 c) 200 d) 255 [4]

5. Convert to denary: a) 00101101 b) 11110000 c) 10000001 [3]

6. Convert 0000 1001 1100 0011 (16-bit) to denary. Show working. [2]

7. Why does 11111111 + 00000001 cause a problem in 8-bit? Name it. [2]

SECTION B: HEXADECIMAL CONVERSIONS (8 marks)

8. Convert to hex: a) 1101 1110 b) 1010 1011 c) 1111 0000 0000 1111 [4]

6. Convert to denary: a) 2A b) FF c) 1B4 [4]

SECTION C: FILE SIZE CALCULATIONS (10 marks)

9. Bitmap 640x480, 16-bit depth. a) File size in KB (show working) [3]

b) Converted to 1-bit B&W: new size? [2]

10. Sound: 44,100Hz, 16-bit, 3 minutes, mono. a) Size in MB [3]

b) At 22,050Hz, 8-bit: new size and trade-off? [2]

SECTION D: APPLICATION (7 marks)

11. Store 500 as 16-bit unsigned: a) Binary b) Hex c) Why range 0-65535? [5]

12. Two ways to reduce image file size WITHOUT compression. [2]

--- WORKSHEET 1B — COMPRESSION, TEXT REPRESENTATION & SOUND -----

Total: 35 marks | Name: _____ Class: _____ Date: _____

SECTION A: COMPRESSION (12 marks)

1. Explain lossy vs lossless with one example each. [4]

2. RLE: X X X X Y Y Z Z Z Z Z A A B B B B B B

a) Compressed form b) Compression ratio c) One image type where RLE effective, one where not. Explain. [8]

13. BMP=2.4MB, JPEG=0.3MB. a) Storage saved b) Why JPEG smaller?

c) Re-saved 10 times: effect on quality? [5]

SECTION B: TEXT REPRESENTATION (8 marks)

14. ASCII: a) How many characters? b) Why not suitable for Chinese? [3]

15. Unicode: a) Max characters? b) Why UTF-8 variable length more efficient?

c) Two advantages over ASCII. [5]

SECTION C: SOUND (8 marks)

16. Describe analog-to-digital conversion (sampling rate, bit depth). [4]

17. Studio: 96,000Hz, 24-bit, stereo. a) Bit rate (kbps) b) Why not CD quality? [4]

SECTION D: MIXED (7 marks)

18. Image: 1024x768, 24-bit. a) Uncompressed MB b) Lossless to 1.5MB: ratio

c) Why lossy achieves greater ratio? [7]

--- ANSWER KEY 1A (35 marks) -----

1a: 00101101 b: 10000000 c: 11001000 d: 11111111 [4]

2a: 45 b: 240 c: 129 [3]

3: $2048 + 128 + 64 + 2 + 1 = 2243$ [2] ([1] method, [1] answer)

4: Result needs 9 bits (1 0000 0000), 9th bit lost. Called overflow. [2]

5a: DE b: AB c: F00F [4]

6a: 42 b: 255 c: 436 [4]

7a: $640 \times 480 \times 16 / 8 / 1024 = 600\text{KB}$ [3] b: $640 \times 480 \times 1 / 8 / 1024 = 37.5\text{KB}$ [2]

8a: $44100 \times 16 \times 180 / 8 / 1048576 = 15.09\text{MB}$ [3] b: 3.77MB, lower quality [2]

9a: 00000001 11110100 [2] b: 01F4 [1] c: $2^{16} = 65536$ values, unsigned = 0-65535 [2]

10: Reduce resolution / reduce colour depth / crop image [2]

--- ANSWER KEY 1B (35 marks) -----

1: Lossy removes data permanently, cannot recover, ex: JPEG/MP3/MP4 [2]

Lossless preserves all data, reconstructs original, ex: PNG/FLAC/ZIP [2]

2a: X4 Y2 Z5 A2 B6 [2] b: 19:10 or 1.9:1 [2]

c: Effective: simple graphics/cartoons (large same-colour areas) [2]

Not effective: photographs (few repeated values) [2]

3a: 2.1MB [1] b: Lossy removes less-sensitive data [2]

c: Each save re-applies lossy, quality degrades [2]

- 4a: 128 [1] b: Only 128 chars, Chinese has thousands [2]
- 5a: $2^{32}=4.3$ billion [1] b: UTF-8: English=1 byte, fixed 4 bytes wastes space[2]
 c: Supports all languages/Chinese/emojis/backward compatible with ASCII [2]
- 6: Sample at intervals (sampling rate), convert to binary (bit depth) [1]
 Higher rate=more samples=better quality [1]
 Higher depth=more precision=better quality [2]
- 7a: $96000 \times 24 \times 2 = 4608000$ bps = 4608 kbps [2]
 b: Captures more detail, allows more editing before audible loss [2]
- 8a: $1024 \times 768 \times 24 / 8 / 1048576 = 2.25$ MB [2] b: $2.25:1.5 = 1.5:1$ [2]
 c: Lossy removes data human vision less sensitive to [3]

WEEK 2: DATA TRANSMISSION, HARDWARE & SOFTWARE

Syllabus Ref: T1-3, T4.1-4.5 | 2 lessons, 2 worksheets, answer keys

--- LESSON 2.1 — DATA TRANSMISSION, ERROR DETECTION & ENCRYPTION -----

Topic: Transmission & Security | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/1 4.1-4.3, 4.5

Learning Objective: Describe packet switching, identify transmission methods, calculate parity, compare symmetric/asymmetric encryption, explain SSL/TLS.

Key Vocabulary:

- packet switching (分组交换 / fenzu jiaohuan) - circuit switching (电路交换)
- router (路由器 / luyouqi) - parity bit (奇偶校验位 / ji'ou jiaoyan wei)
- checksum (校验和 / jiaoyan he) - symmetric encryption (对称加密 / duichen jiami)
- asymmetric encryption (非对称加密 / feiduichen jiami) - SSL/TLS
- serial (串行传输 / chuanxing chuanshu) - parallel (并行传输 / bingxing chuanshu)
- simplex (单工 / dangong) - half-duplex (半双工 / banshuanggong)
- full-duplex (全双工 / quanshuanggong)

Lesson Structure (40 min):

0-3 | DO NOW: "List ways data can travel between two computers."

3-12 | DIRECT INSTRUCTION:

19. Packet Switching: data->packets (header: source/dest IP, seq#, payload, trailer). Routers forward independently by IP address.
 At destination: reassembled by sequence#. Advantages vs circuit:
 more efficient use of network, resilient (rerouting), shared,
 no setup time.
20. Transmission: Serial (1 bit, 1 wire, slower, long distance: USB, Ethernet). Parallel (multiple bits, multiple wires, faster,

expensive, signal skew: CPU-RAM buses). Simplex (one direction).

Half-duplex (both, not simultaneous). Full-duplex (both simultaneously).

21. Error Detection: Parity (add bit for even/odd count of 1s.

Limitation: even flips undetected). Checksum (sum data bytes, compare at receiver. Used in TCP/IP headers).

22. Encryption: Symmetric (same key, fast, key distribution problem).

Asymmetric (public encrypt, private decrypt, slower, solves key distribution). SSL/TLS (asymmetric for session key exchange, symmetric for data transfer). The "S" in HTTPS.

12-25 | GUIDED PRACTICE: Packet switching simulation (5 students as routers, paper packets with seq#). Parity calculations in pairs.

25-35 | INDEPENDENT PRACTICE: Worksheet 2A.

35-40 | PLENARY: Oral quiz. HW: Complete Worksheet 2A. Revise CPU for L2.2.

EAL Support: Network diagram (bilingual), kinaesthetic simulation, mnemonic "SS PP HH FF", simplex=one-way street (单行道).

Differentiation:

Emerging: Match terms, guided parity

Developing: Explain packet switching, parity independently, compare encryption

Secure: Analyse resilience, calculate checksum, explain SSL/TLS handshake

Resources: Slides, paper packets, router role cards, Worksheet 2A.

--- LESSON 2.2 — CPU, MEMORY, STORAGE, I/O & SOFTWARE -----

Topic: Computer Architecture & Software | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/1 1.1-1.4, 2.1-2.3

Learning Objective: Draw Von Neumann with all registers, explain FDE with trace table, compare storage types, state OS functions, compare translators.

Key Vocabulary:

- CPU (中央处理器 / zhongyang chuli qi) - ALU (算术逻辑单元)
- CU (控制单元 / kongzhi danyuan) - register (寄存器 / jicunqi)
- PC (程序计数器 / chengxu jishuqi) - MAR/MDR/CIR/ACC
- cache (高速缓存 / gaosu huancun) - RAM/ROM
- virtual memory (虚拟内存 / xuni neicun) - interrupt (中断 / zhongduan)
- compiler (编译器 / bianyi qi) - interpreter (解释器 / jieshi qi)
- assembler (汇编器 / huibian qi)

Lesson Structure (40 min):

0-3 | DO NOW: Draw Von Neumann from memory.

3-12 | DIRECT INSTRUCTION:

23. Von Neumann: CPU(CU+ALU+registers), Memory(RAM/ROM), I/O, Buses.

REGISTERS: PC=next instruction address, CIR=current instruction,

MAR=memory address accessed, MDR=data to/from memory,

ACC=calculation results.

24. FDE Cycle: FETCH: PC->MAR, address bus->memory, data->MDR->CIR,

PC increments. DECODE: CU decodes opcode in CIR. EXECUTE: CU

signals ALU, result to ACC or memory.

25. Memory Hierarchy: Registers->Cache(L1,L2)->RAM->Virtual Memory.

Cache: small, fast, frequent data. Virtual memory: HDD/SSD as

extra RAM, slower, thrashing if overused.

26. Storage: Magnetic (HDD/tape: large, cheap, slow, moving parts).

Optical (CD/DVD/BD: removable, cheap, slow, fragile).

Solid-state (SSD/USB: fast, durable, silent, expensive/GB).

27. I/O: Sensors (temperature, pressure, light, moisture, pH, motion,

proximity, gas, sound, infrared). Actuators (motor, pump, heater,

buzzer, LED, relay, servo). Embedded systems: dedicated computer

in device (washing machine, car ECU, microwave).

28. OS Functions: memory management, processor scheduling, device

management, file management, security, user interface.

Interrupts: signal stops processing for urgent event. Process:

save state -> run ISR -> restore -> resume.

29. Translators: Compiler (entire source->machine code, executable,

all errors after). Interpreter (line-by-line, stops at first error,

slower, easier debugging). Assembler (assembly->machine code 1:1,

mnemonics LDA/ADD/STO).

12-25| GUIDED PRACTICE: FDE trace table (LDA 150, ADD 151, STO 152).

Storage device selection for 5 scenarios.

25-35| INDEPENDENT PRACTICE: Worksheet 2B.

35-40| PLENARY: Rapid recall game. HW: Complete Worksheet 2B. CPU register cards.

EAL Support: Large labelled diagram (bilingual), colour-coded (blue=CPU,

green=memory, red=buses), mnemonic: "PC points, CIR carries, MAR marks

address, MDR moves data, ACC accumulates", kinaesthetic FDE roleplay.

Differentiation:

Emerging: Label diagrams, match definitions, 2-instruction FDE only

Developing: Draw from memory, FDE traces, explain hierarchy, compare translators

Secure: 4+ instruction traces, storage cost analysis, cache locality,

interrupt priority analysis

Resources: Whiteboard diagram, FDE templates, storage samples, Worksheet 2B.

--- WORKSHEET 2A — DATA TRANSMISSION, ERROR DETECTION & ENCRYPTION -----

Total: 35 marks | Name: _____ Class: _____ Date: _____

SECTION A: PACKET SWITCHING (10 marks)

30. Describe packet switching (sender, network, receiver). [5]

31. Compare to circuit switching. Two advantages of packet. [3]

3. Packet header has seq# 7 of 12. Why needed? [2]

SECTION B: TRANSMISSION METHODS (6 marks)

32. State method + reason: a) Computer to nearby printer b) Video call

c) Keyboard to computer [6]

SECTION C: ERROR DETECTION (8 marks)

33. Even parity: 1101 011_ a) Calculate parity bit b) Bits 3&5 flipped:

receiver parity? Error detected? Explain. [5]

6. Describe checksum. [3]

SECTION D: ENCRYPTION (11 marks)

7. Explain symmetric vs asymmetric encryption. [4]

34. Alice->Bob (asymmetric): a) Which key encrypts? b) Which decrypts?

c) Why asymmetric slower? [4]

9. Explain SSL/TLS. [3]

--- WORKSHEET 2B — CPU, MEMORY, STORAGE, I/O & SOFTWARE -----

Total: 40 marks | Name: _____ Class: _____ Date: _____

SECTION A: CPU (10 marks)

1. Draw labelled Von Neumann (CPU, memory, I/O, buses). [4]

2. State purpose: a) PC b) MAR c) MDR d) CIR e) ACC [5]

SECTION B: FDE CYCLE (10 marks)

35. Trace: Addr100=LDA150, 101=ADD151, 102=STO152. Mem[150]=12, Mem[151]=8.

Complete 6-step trace showing PC/MAR/MDR/CIR/ACC. [6]

4. Role of CU in FDE. [2] 5. Role of ALU. [2]

SECTION C: MEMORY & STORAGE (10 marks)

36. Order: Virtual Memory, Cache, RAM, Registers (fastest to slowest). [2]

7. Purpose of cache. [2]

37. Compare magnetic/optical/solid-state: example+advantage+disadvantage each.[6]

SECTION D: SOFTWARE (10 marks)

38. Four OS functions. [4] 10. What happens when interrupt occurs. [3]

39. Compare compiler and interpreter. One advantage each. [3]

--- ANSWER KEY 2A (35 marks) -----

1: Split into packets[1], header with addresses/seq#[1], travel independently[1], routers forward[1], reassembled at dest[1]. [5]

2: Circuit=dedicated connection[1], packet=no dedicated connection[1].

Advantages: efficient/resilient/shared/no setup [2]. [3]

3: Packets arrive out of order[1], seq# reassembles correctly[1]. [2]

4a: Parallel (short distance, fast)[2] b: Full-duplex (both directions)[2]

c: Serial/Simplex (one direction)[2]

5a: Five 1s (odd), parity=1 [2]

b: 1000 1111, eight 1s (even), NOT detected (even flips) [3]

6: Calculate sum from data bytes[1], send with data[1], receiver recalculates[1].

7: Symmetric=same key[1], both need key[1]. Asymmetric=public encrypt[1], private decrypt, public shared openly[1].

8a: Bob's public key[1] b: Bob's private key[1]

c: Complex modular arithmetic with large primes[2].

9: Asymmetric for session key exchange[1], symmetric for data[1], combines security+speed[1].

--- ANSWER KEY 2B (40 marks) -----

1: CPU with CU/ALU/registers[1], memory[1], I/O[1], buses[1].

2a: PC=address of next instruction b: MAR=memory address accessed

c: MDR=data to/from memory d: CIR=current instruction

e: ACC=calculation result [5]

3: Six rows, [1] each. Row2:PC101,MAR150,MDR12,ACC12. Row4:PC102,MAR151,ACC20. Row6:PC103,MAR152,MDR20.[6]

4: Decodes instructions[1], sends control signals[1].

5: Arithmetic and logical operations[2].

6: Registers, Cache, RAM, Virtual Memory[2].

7: Stores frequent data near CPU[1], reduces waiting[1].

8: Magnetic:HDD/tape[0.5],large capacity/low cost[1.5]. Optical:CD/DVD[0.5],

- removable/cheap[1.5]. Solid-state:SSD/USB[0.5],fast/no moving parts[1.5].
- 9: Memory/processor/device/file/security/user interface management [4]
- 10: Save state[1], run ISR[1], restore and resume[1].
- 11: Compiler=entire program at once[1], Interpreter=line by line[1].
- Compiler:faster[1], Interpreter:easier debugging[1].

WEEK 3: INTERNET, SECURITY & EMERGING TECHNOLOGIES

Syllabus Ref: T4, T5, T6 | 2 lessons, 2 worksheets, answer keys

--- LESSON 3.1 — INTERNET, NETWORKS & DIGITAL CURRENCY -----

Topic: Internet Technologies | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/1 4.1, 4.2, 4.4, 5.1

Learning Objective: Distinguish Internet/WWW, explain DNS, compare HTTP/HTTPS, describe routers/IP addresses, explain blockchain, compare digital currencies.

Key Vocabulary:

- Internet (互联网 / Hulianwang) - WWW (万维网 / Wanweiwang)
- URL (统一资源定位符 / Tongyi ziyuan dingwei fu) - DNS (域名系统 / Yuming xitong)
- IP address (IP 地址) - IPv4/IPv6 - router (路由器 / luyouqi)
- MAC address (媒体访问控制地址) - NIC (网卡 / wangka)
- blockchain (区块链 / qukuai lian) - digital currency (数字货币 / shuzi huobi)
- e-CNY (数字人民币 / shuzi renminbi) - Bitcoin (比特币 / bitebi)

Lesson Structure (40 min):

0-3 | DO NOW: "Type www.google.com - what happens? List all steps."

3-12 | DIRECT INSTRUCTION:

40. Internet vs WWW: Internet=global network of networks (TCP/IP).
WWW=service ON the Internet (HTTP/HTTPS). Analogy: Internet=roads,
WWW=traffic.
41. URL Structure: protocol://domain:port/path?query#fragment
https://www.example.com:443/path/page.html?id=1
42. DNS Lookup: (1) Browser checks cache (2) Query local DNS (ISP)
(3) Root DNS -> TLD server (.com) (4) TLD -> authoritative DNS
(5) Authoritative returns IP (6) IP cached locally (7) Browser connects.
DNS uses port 53.
43. HTTP vs HTTPS: HTTP=port 80, unencrypted. HTTPS=port 443,
encrypted (SSL/TLS), authenticates server.
44. Routers: connect networks, read dest IP, consult routing table,
forward to next hop. NAT: multiple LAN devices share one public IP.

45. IP Addresses: IPv4=32-bit, 4.3 billion, running out.

IPv6=128-bit, 340 undecillion. Private: 10.x, 172.16-31.x, 192.168.x.

46. Network Hardware: NIC (connects device, unique MAC 48-bit hex).

Switch (same LAN, MAC addresses). Router (different networks, IP).

Modem (digital<->analog). Access Point (wireless to wired).

47. Digital Currency: Blockchain=distributed ledger, chain of blocks, decentralised, cryptographic hashing. e-CNY (PBOC centralised CBDC, legal tender, stable). Bitcoin (decentralised, limited 21M, volatile, proof-of-work mining). Comparison: e-CNY=centralised/stable/legal; Bitcoin=decentralised/volatile/not legal tender.

12-25| GUIDED PRACTICE: DNS lookup roleplay (7 students: user, cache, local DNS, root, TLD, authoritative, web server). IP classification exercise.

25-35| INDEPENDENT PRACTICE: Worksheet 3A.

35-40| PLENARY: "Explain Internet vs WWW in one sentence." HW: Worksheet 3A.

Revise malware for L3.2.

EAL Support: Visual DNS hierarchy diagram, step-by-step DNS flowchart (laminated), real-world analogy (DNS=phone book), e-CNY personal relevance for Chinese students.

Differentiation:

Emerging: Match terms, label URL parts, identify IPv4/IPv6

Developing: Explain DNS step-by-step, compare HTTP/HTTPS, explain router, describe blockchain

Secure: Full DNS with cache hierarchy, IPv4/IPv6 detail, analyse e-CNY vs Bitcoin with centralisation arguments, explain NAT

Resources: Slides, DNS roleplay cards, IP cards, Worksheet 3A.

--- LESSON 3.2 — CYBERSECURITY, EMERGING TECH & ETHICS -----

Topic: Security & Emerging Technologies | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/1 5.1-5.3, 6.1-6.3

Learning Objective: Name/describe all malware types, list all protections, describe automated systems/robotics/AI, discuss ethical considerations.

Key Vocabulary:

- malware (恶意软件 / eyi ruanjian) - virus (病毒 / bingdu) - worm (蠕虫 / ruchong)
- trojan (木马 / muma) - ransomware (勒索软件 / lesuo ruanjian)
- spyware (间谍软件 / jiandie ruanjian) - adware (广告软件 / guanggao ruanjian)
- phishing (网络钓鱼 / wangluo diaoyu) - firewall (防火墙 / fanghuoqiang)
- antivirus (杀毒软件 / shadu ruanjian) - 2FA (双因素认证 / shuang yinsu renzheng)
- automated system (自动化系统 / zidonghua xitong) - robotics (机器人技术)

- AI (人工智能 / rengong zhineng) - machine learning (机器学习 / jiqi xuexi)
- ethics (伦理 / lunli) - privacy (隐私 / yinsi)

Lesson Structure (40 min):

0-3 | DO NOW: "Name as many malware types as you can. What does each do?"

3-12 | DIRECT INSTRUCTION:

48. MALWARE: Virus (attaches to programs, needs user action, corrupts data). Worm (self-replicating, spreads independently, consumes bandwidth). Trojan (disguised as legitimate, user installs willingly, creates backdoor, doesn't self-replicate). Ransomware (encrypts files, demands payment: WannaCry, NotPetya). Spyware (monitors secretly, keylogging, steals passwords). Adware (unwanted ads, bundled with free software). Phishing (social engineering via fake emails/sites - delivery method, not malware itself).
 49. PROTECTIONS: Firewall (monitors traffic, blocks unauthorised access). Antivirus (detects/removes malware, signature+heuristic, must update). Encryption (scrambles data, protects in transit and at rest). 2FA (two forms of ID: know+have/are). Backup (3-2-1 rule: 3 copies, 2 media, 1 offsite - ONLY protection against ransomware). Patches (fix vulnerabilities - WannaCry spread via unpatched Windows). User Access Levels (principle of least privilege).
 50. AUTOMATED SYSTEMS: Sensors->microprocessor->actuators. Examples: traffic lights, automatic doors, washing machines, temperature control. Advantages: consistent, 24/7, faster, safer. Disadvantages: job displacement, high cost, inflexible.
 51. ROBOTICS: Physical machines with sensors, processors, actuators. Industrial: assembly, welding, painting. Service: vacuum, surgical, exploration. Components: manipulator, end effector, sensors, controller.
 52. AI: Machines simulating human intelligence. ML: learns from data without explicit programming. Narrow AI: one task. General AI: human-level (not yet achieved). Applications: voice assistants, image rec, recommendations, autonomous vehicles, medical diagnosis.
 53. ETHICS: Job displacement, privacy/data collection, bias in AI, accountability, dependence, digital divide, environmental cost (AI training energy, crypto mining).
- 12-25 | GUIDED PRACTICE: Malware-protection matching. Hospital ransomware scenario analysis (what protections needed? what to do? ethics?).

25-35 | INDEPENDENT PRACTICE: Worksheet 3B.

35-40 | PLENARY: "List all 6 malware types and one protection each."

HW: Worksheet 3B. Begin revision for Paper 1 timed practice (W4).

EAL Support: Colour-coded malware icons, Chinese cultural connection (e-CNY, QR payments, WannaCry affected China), mnemonic "VWTRSA".

Differentiation:

Emerging: Match malware to descriptions, list protections, basic definitions

Developing: Describe malware in own words, explain layered security, give automated system examples, discuss one ethical issue

Secure: Compare malware similarities/differences, evaluate protection combos, analyse ethical dilemmas, explain ML vs traditional programming

Resources: Slides, malware icon cards, hospital scenario, Worksheet 3B.

--- WORKSHEET 3A — INTERNET, NETWORKS & DIGITAL CURRENCY -----

Total: 35 marks | Name: _____ Class: ____ Date: _____

SECTION A: INTERNET & WWW (8 marks)

1. Explain Internet vs WWW. [3]

2. Describe DNS lookup process (at least 4 steps). [5]

SECTION B: URLS, PROTOCOLS & IP (10 marks)

54. Break down: <https://www.school.edu.cn:443/students/login.php?id=123>

Identify: protocol, domain, port, path, query. [5]

4. Two differences between HTTP and HTTPS. [2]

5. Two advantages of IPv6 over IPv4. [3]

SECTION C: NETWORK HARDWARE (7 marks)

6. Describe function: a) Router b) Switch c) NIC

d) Difference between MAC address and IP address. [7]

SECTION D: DIGITAL CURRENCY (10 marks)

7. Explain blockchain. [4]

8. Compare e-CNY with Bitcoin (two differences). [4]

55. One advantage and one disadvantage of digital currencies. [2]

Total: 40 marks | Name: _____ Class: ____ Date: _____

SECTION A: MALWARE (12 marks)

1. Describe what each does and how it spreads:

a) Virus b) Worm c) Trojan d) Ransomware [8]

2. What is phishing? Why effective? [2]

56. Email attachment "invoice.pdf.exe" - two reasons not to open. [2]

SECTION B: PROTECTIONS (12 marks)

4. Describe: a) Firewall b) Antivirus c) 2FA d) Patches [8]

57. Explain 3-2-1 backup rule. Why only effective protection vs ransomware?[4]

SECTION C: EMERGING TECH (8 marks)

58. Automated greenhouse: a) Name sensor and actuator b) How system works [4]

7. Difference between robotics and AI. [2]

8. One machine learning application. [2]

SECTION D: ETHICS (8 marks)

9. AI screens job apps, trained on mostly male data.

a) Ethical problem b) Why occurs c) Suggest fix [4]

10. Two ethical concerns about factory automation. [4]

--- ANSWER KEY 3A (35 marks) -----

1: Internet=global network of networks[1], uses TCP/IP[0.5].

WWW=service on Internet[1], uses HTTP/HTTPS[0.5]. [3]

2: Browser checks cache[1], query local DNS[1], root->TLD->authoritative[1],

returns IP, browser connects[1], IP cached[0.5]. [5]

3: Protocol:https[1] Domain:www.school.edu.cn[1] Port:443[1]

Path:/students/login.php[1] Query:id=123[1]. [5]

4: HTTPS encrypted HTTP not[1]; HTTPS port 443 HTTP 80[1]; HTTPS authenticates[1]

(any two). [2]

5: 128-bit vs 32-bit (more addresses)[1]; built-in security[1];

efficient routing[1]; no NAT needed[1] (any two). [3]

- 6a: Router connects networks, forwards by IP[2]. b: Switch same LAN, MAC[2].
 c: NIC connects device to network[1]. d: MAC=hardware ID(permanent),
 IP=logical(changable)[2]. [7]
- 7: Distributed digital ledger[1], transactions in blocks[1], cryptographically
 linked by hashes[1], distributed across nodes, tamper-resistant[1]. [4]
- 8: Any two: e-CNY centralised(PBOC) vs Bitcoin decentralised[2];
 e-CNY legal tender vs Bitcoin not[2]; e-CNY stable vs Bitcoin volatile[2];
 e-CNY no mining vs Bitcoin proof-of-work[2]. [4]
- 9: Advantage: faster/lower fees/24-7[1]. Disadvantage: needs internet/
 security risks/privacy[1]. [2]
- ANSWER KEY 3B (40 marks) -----
- 1a: Virus=attaches to programs, needs user action, corrupts data[2]
 b: Worm=self-replicating, spreads across networks, no user action[2]
 c: Trojan=disguised as legitimate, user installs, creates backdoor[2]
 d: Ransomware=encrypts files, demands payment[2]. [8]
- 2: Fake emails/websites appearing from trusted sources[1], tricks users into
 revealing sensitive info[1]. [2]
- 3: Extension is .exe not .pdf (executable)[1]; executables install malware[1].[2]
- 4a: Firewall=monitors traffic, blocks unauthorised access[2]
 b: Antivirus=detects/removes malware, signatures+heuristics, must update[2]
 c: 2FA=two forms of ID (know+have/are)[2]
 d: Patches=fix security vulnerabilities[2]. [8]
- 5: 3 copies, 2 media types, 1 offsite[2]. Only protection because once
 encrypted, only uncompromised copy allows recovery[2]. [4]
- 6a: Sensor:temperature/light/moisture[1] Actuator:heater/fan/motor[1]
 b: Sensor measures[1], microprocessor compares, activates actuator[1]. [4]
- 7: Robotics=physical machines doing physical tasks[1]; AI=simulating human
 intelligence, may not have physical body[1]. [2]
- 8: Any valid: recommendations/voice recognition/image recognition/medical[2].[2]
- 9a: Gender bias/discrimination[1] b: Learned from biased historical data[2]
 c: Retrain on balanced data/remove gender data/manual review[1]. [4]
- 10: Job displacement/economic hardship[2]; Loss of human skills[2];
 Safety concerns[2]; Economic inequality[2] (any two with explanation). [4]

WEEK 4: PAPER 1 TIMED PRACTICE

Syllabus Ref: T1-6 (Complete Paper 1 revision) | 2 lessons, timed paper

--- LESSON 4.1 — TIMED PRACTICE EXAM (40-MINUTE SECTION) -----

Topic: Paper 1 Mixed Topics | Duration: 40 min | Lesson Type: Exam Practice

Syllabus Ref: 0478/1 All T1-6

Learning Objective: Practice under timed exam conditions, apply revision knowledge, experience time-limited assessment pressure.

Key Vocabulary: exam conditions (考试条件 / kaoshi tiaojian),

time management (时间管理 / shijian guanli), show your working (展示演算)

Lesson Structure (40 min):

0-3 | DO NOW: EXAM TECHNIQUE REMINDERS on board:

Read Q twice before answering. Check marks=more detail. 1-mark=one fact.

2-mark=two points. 3-4 marks=explain with examples. 6+=paragraphs.

If stuck: move on, come back. Show ALL working. Write clearly.

3-5 | DIRECT INSTRUCTION: "40 minutes ~28 marks ~1.4 min/mark. No talking, no phones. Manage your own time. 10-min warning at 25 min."

5-35 | TIMED PRACTICE: Students complete Timed Practice Paper 1.

Teacher invigilates. 10-min warning at 25 min.

35-40 | COLLECTION + PLENARY: "Which was hardest?" Thumbs up/down.

HW: Relax - review in L4.2.

EAL Support: Extra 5 min if needed. Key vocabulary list on request.

Differentiation:

Emerging: Target 50%, focus on 1-2 mark questions, skip 6-mark if needed

Developing: Target 75%, attempt all, proportionate time per question

Secure: Complete all with time to check, structured 6-mark answers

Resources: Timed Practice Paper 1 (printed), extra paper, visible clock.

--- LESSON 4.2 — REVIEW, MARK SCHEME & EXAM TECHNIQUE -----

Topic: Paper 1 Review | Duration: 40 min | Lesson Type: Review

Syllabus Ref: 0478/1 All T1-6

Learning Objective: Self-mark using mark scheme, identify gaps, create targeted revision plan.

Key Vocabulary: self-assessment (自我评估 / ziwo pinggu),

mark scheme (评分标准 / pingfen biao zhun), gap analysis (差距分析 / chaju fenxi)

Lesson Structure (40 min):

0-3 | DO NOW: "Without looking at paper, write 3 things done well, 3 difficult."

3-20 | MARK SCHEME WALKTHROUGH: Project mark scheme. Each Q: read, show marks,

"Who got full?", common errors. Key errors: padding zeros, explain vs

define, compare one-sided only, forgetting /8 or /1024, MAR/MDR confusion.

20-30| SELF-MARKING: Green pen. Total at top. Teacher circulates.

30-35| GAP ANALYSIS: Topic breakdown table with % and action needed.

35-40| PLENARY: "Top 3 areas to revise before Mock 2."

HW: Revise weakest topic.

EAL Support: Bilingual gap analysis form, individual target discussion, buddy peer revision.

Differentiation:

Emerging: Teacher-assisted marking, simplified analysis, specific targets

Developing: Self-mark, full gap analysis, own revision targets

Secure: Self-mark, full analysis, create revision resources for peers

--- TIMED PRACTICE PAPER 1 (40 MINUTES, 28 MARKS) -----

Name: _____ Class: _____ Date: _____ Time: 40 min Total: 28

Instructions: Answer ALL. Show all working. No calculators.

SECTION A: DATA REPRESENTATION (8 marks)

1. Convert 183 to 8-bit binary. [1]

2. Convert 1100 1110 to hex. [1]

3. Add: 0110 1101 + 0011 0101. Show working. [2]

59. Image: 320x240, 8-bit depth. a) File size in KB (show working) [2]

b) Compressed lossless to 45KB: compression ratio [1]

c) Why would lossy achieve greater ratio than lossless? [1]

SECTION B: HARDWARE & SOFTWARE (8 marks)

5. Describe roles of CU and ALU. [2]

60. Complete FDE trace table: 100=LDA150, 101=ADD151, 102=STO152.

Mem[150]=25, Mem[151]=17. Show PC/MAR/MDR/CIR/ACC for 6 cycles. [4]

7. One difference between RAM and ROM. [2]

SECTION C: NETWORKS & SECURITY (8 marks)

8. Describe packet switching. [3]

61. Student connects via HTTPS. a) Two advantages over HTTP [2] b) Port? [1]

10. Name and describe two malware types. [2]

SECTION D: EMERGING TECH (4 marks)

62. Automatic greenhouse: a) Sensor and actuator [2] b) How system works [2]

--- MARK SCHEME — TIMED PRACTICE PAPER 1 (28 marks) -----

1: 10110111 [1]

2: CE [1]

3: 1010 0010 [2] (working[1], answer[1])

4a: $320 \times 240 \times 8 / 8 / 1024 = 75\text{KB}$ [2] b: $75:45=5:3$ [1]

c: Lossy permanently removes less-sensitive data [1]

5: CU decodes instructions/sends control signals[1]; ALU arithmetic+logical[1]

6: Row1:PC100,MAR100,MDR"LDA150",CIR"LDA150",ACC--

Row2:PC101,MAR150,MDR25,CIR"LDA150",ACC25

Row3:PC101,MAR101,MDR"ADD151",CIR"ADD151",ACC25

Row4:PC102,MAR151,MDR17,CIR"ADD151",ACC42

Row5:PC102,MAR102,MDR"STO152",CIR"STO152",ACC42

Row6:PC103,MAR152,MDR42,CIR"STO152",ACC42 [4] ([1] per 2 rows)

7: RAM volatile read/write[1]; ROM non-volatile read-only[1]

8: Split into packets[1], header with addresses[1], travel independently,
reassembled at dest[1]. [3]

9a: HTTPS encrypted authenticates[1]; HTTPS port 443[1] b: 443[1]

10: Any two malware with descriptions [2]

11a: Any valid sensor+actuator [2] b: Sensor measures[1], microprocessor
compares, activates actuator[1]. [2]

--- GAP ANALYSIS TEMPLATE — PAPER 1 TIMED PRACTICE -----

Name: _____ Date: _____ Score: ____/28

Topic	Q	Avail	Score	%	Priority
Binary/Hex	1,2	2			
Binary Addition	3	2			
File Size	4	4			
CPU Architecture	5	2			
FDE Cycle	6	4			
RAM/ROM	7	2			
Packet Switching	8	3			
HTTPS/HTTP	9	3			

Malware	10	2			
Automated Systems	11	4			

TOP 3 PRIORITIES: 1: _____ 2: _____ 3: _____

ACTION PLAN: Re-read notes / Re-attempt worksheets / Ask teacher / Flashcards

UNIT 2: PAPER 2 INTENSIVE REVISION

#

WEEK 5: ALGORITHMS & PROBLEM SOLVING

Syllabus Ref: T7 (Algorithm design) | 2 lessons, 2 worksheets, answer keys

--- LESSON 5.1 — PSEUDOCODE, FLOWCHARTS & PROBLEM SOLVING -----

Topic: Algorithm Design | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/2 7.1-7.4

Learning Objective: Write Cambridge pseudocode, draw/interpret flowcharts, apply decomposition and abstraction, convert between pseudocode/flowcharts.

Key Vocabulary: pseudocode (伪代码 / wei daima), flowchart (流程图 / liucheng tu),

algorithm (算法 / suanfa), decomposition (问题分解 / wenti fenjie),

abstraction (抽象 / chouxang), iteration (迭代 / diedai),

selection (选择 / xuanze), sequence (顺序 / shunxu)

Lesson Structure (40 min):

0-3 | DO NOW: "Write pseudocode: input a number, output if positive/negative/zero."

3-12 | DIRECT INSTRUCTION:

CAMBRIDGE PSEUDOCODE CONSTRUCTS (all must be known):

DECLARE Counter : INTEGER

Counter <- 0 (assignment arrow)

INPUT UserName / OUTPUT "Hello"

IF Age >= 18 THEN ... ELSE ... ENDIF

FOR i <- 1 TO 10 ... ENDFOR

FOR i <- 10 DOWNT0 1 ... ENDFOR

WHILE Counter <= 100 DO ... ENDWHILE (pre-test, may run 0 times)

REPEAT ... UNTIL Password = "secret" (post-test, runs at least once)

FLOWCHART SYMBOLS: Oval=start/end, Parallelogram=input/output,

Rectangle=process, Diamond=decision, Arrow=flow.

DECOMPOSITION: Break complex problem into sub-problems (e.g. game:

character design, level design, scoring, graphics). Advantages: easier

to understand, different programmers, easier testing.

ABSTRACTION: Remove unnecessary detail (e.g. map shows roads not every building). Using functions to hide complex details.

12-25| GUIDED PRACTICE: Model pseudocode for "Input 5 numbers, output average."

Draw flowchart for login system in pairs.

25-35| INDEPENDENT PRACTICE: Worksheet 5A.

35-40| PLENARY: "Convert flowchart to pseudocode" (teacher draws on board).

HW: Complete Worksheet 5A.

EAL Support: Laminated pseudocode reference (bilingual), flowchart icons, scaffolded fill-in-blank before full writing, colour-coded constructs.

Differentiation:

Emerging: Fill-in-blank, 3 basic constructs (assignment, IF, FOR)

Developing: Write independently, convert pseudocode $\leftrightarrow$ flowcharts, explain decomposition/abstraction with examples

Secure: Efficient pseudocode, complex conversions, critique inefficient algos

Resources: Slides, laminated construct sheets, Worksheet 5A.

--- LESSON 5.2 — SEARCHING, SORTING & TRACE TABLES -----

Topic: Algorithms & Efficiency | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/2 7.5-7.7

Learning Objective: Trace linear/binary search, compare efficiency, trace bubble sort, state $O(n^2)$, complete trace tables.

Key Vocabulary: linear search (线性搜索 / xianxing sousuo),

binary search (二分搜索 / erfen sousuo), bubble sort (冒泡排序 / maopao paixu),

trace table (跟踪表 / genzong biao), time complexity (时间复杂度 / shijian fuza du),

sorted (已排序 / yi paixu), unsorted (未排序 / wei paixu)

Lesson Structure (40 min):

0-3 | DO NOW: "1000 items. Worst case comparisons: linear? binary?"

(Answers: 1000, ~10)

3-12 | DIRECT INSTRUCTION:

LINEAR SEARCH: Check each item in order. Best: $O(1)$, Worst: $O(n)$,

Average: $n/2$. Works on ANY list (sorted or unsorted).

BINARY SEARCH: Check middle, if target larger search right half, smaller search left. Best: $O(1)$, Worst: $O(\log n)$. List MUST be sorted.

For 1,000,000 items: linear=1,000,000 comparisons, binary=~20.

BUBBLE SORT: Repeated passes, compare adjacent, swap if wrong order.

Largest "bubbles" to end. After each pass, one more item correct.

Example [5,3,8,2]: Pass1:[3,5,2,8], Pass2:[3,2,5,8], Pass3:[2,3,5,8].

Time complexity: $O(n^2)$. For 100 items: ~10,000 comparisons.

TRACE TABLES: Grid showing variable values at each step.

12-25| GUIDED PRACTICE: Search race (20-item sorted list). Bubble sort trace on [9,2,7,4,1].

25-35| INDEPENDENT PRACTICE: Worksheet 5B.

35-40| PLENARY: "30 seconds: tell partner when to use linear vs binary."

HW: Complete Worksheet 5B. Memorise bubble sort pseudocode.

EAL Support: Physical search demonstration (students stand with numbers), pre-printed trace table templates, "bubble" analogy, bilingual comparison table.

Differentiation:

Emerging: Trace linear only, pre-filled tables, state definitions

Developing: Trace both searches, bubble sort traces, compare efficiency, write trace tables from pseudocode

Secure: Write all three algorithms from memory, exact comparison counts, explain $O(n^2)$ mathematically, trace complex nested loops

Resources: Number cards, trace table templates, Slides, Worksheet 5B.

--- WORKSHEET 5A — PSEUDOCODE, FLOWCHARTS & PROBLEM SOLVING -----

Total: 35 marks | Name: _____ Class: ____ Date: _____

SECTION A: PSEUDOCODE WRITING (15 marks)

63. Write pseudocode to: a) Input temperature. If >30 output "Hot", else "OK". [3]

b) Input 10 scores. Count passes (≥ 50) and fails. Output both. [5]

c) Ask for password until "1234" entered. Count attempts. Output count. [4]

d) Output first 10 multiples of 7. [3]

SECTION B: FLOWCHART TO PSEUDOCODE (8 marks)

64. Flowchart: Start $\rightarrow$ Input Number $\rightarrow$ Number $> 0?$ $\rightarrow$ YES: "Positive" $\rightarrow$ End

NO: Number $< 0?$ $\rightarrow$ YES: "Negative" $\rightarrow$ End NO: "Zero" $\rightarrow$ End. Write pseudocode. [4]

65. Draw flowchart for: DECLARE Total,i,Num:INTEGER; Total $\leftarrow$ 0;

FOR i $\leftarrow$ 1 TO 5: INPUT Num; IF Num $>$ 0 THEN Total $\leftarrow$ Total+Num; ENDFOR;

OUTPUT Total [4]

SECTION C: PROBLEM SOLVING (12 marks)

66. Exam results program: input names+scores, calculate grades

(A:80+,B:70+,C:60+,D:50+,F: $<$ 50), output list.

a) How decomposition used? [3] b) Example of abstraction [2]

c) One advantage of decomposition [2]

5. Weather station with multiple sensors every minute.

a) How abstraction simplifies design [3] b) How decomposition used [2]

--- WORKSHEET 5B — SEARCHING, SORTING & TRACE TABLES -----

Total: 35 marks | Name: _____ Class: ____ Date: _____

SECTION A: LINEAR SEARCH (6 marks)

67. Trace linear search for 42 in [15,23,8,42,11,37,56]. Show each comparison. [3]

2. Worst-case comparisons for 500 items? Explain. [2]

3. One advantage of linear over binary. [1]

SECTION B: BINARY SEARCH (8 marks)

68. Trace binary search for 31 in [5,12,18,23,31,42,56,67,89,95]. [4]

5. Trace binary search for 100 in same list. [3]

6. Max comparisons for 1,048,576 items? [1]

SECTION C: BUBBLE SORT (10 marks)

69. Trace bubble sort on [7,2,9,1,5]. Show list after each pass. [5]

8. Write bubble sort pseudocode. [4]

9. Why bubble sort is $O(n^2)$? [1]

SECTION D: TRACE TABLES (11 marks)

70. Trace: A=2,B=3; C<-0; FOR i<-1 TO B: C<-C+A; ENDFOR; OUTPUT C [4]

11. Trace: X=10; REPEAT X<-X-3; OUTPUT X; UNTIL X<=0 [3]

71. Trace: N=6; Factorial<-1; i<-N; WHILE i>0: Factorial<-Factorial*i;
i<-i-1; ENDWHILE; OUTPUT Factorial [4]

--- ANSWER KEY 5A (35 marks) -----

1a: INPUT Temp; IF Temp>30 THEN OUTPUT "Hot" ELSE OUTPUT "OK" ENDIF [3]

b: Pass<-0,Fail<-0; FOR i<-1 TO 10: INPUT Score;

IF Score>=50 THEN Pass<-Pass+1 ELSE Fail<-Fail+1; ENDFOR; OUTPUT Pass,Fail[5]

c: Attempts<-0; REPEAT INPUT Password; Attempts<-Attempts+1;

UNTIL Password="1234"; OUTPUT Attempts [4]

d: FOR i<-1 TO 10: OUTPUT i*7; ENDFOR [3]

2: INPUT Number; IF Number>0 THEN OUTPUT "Positive"

ELSE IF Number<0 THEN OUTPUT "Negative" ELSE OUTPUT "Zero" ENDIF [4]

3: Flowchart with: Oval(Start), Input 3 vars, Process Total<-0, Loop FOR,
Input Num, Decision Num>0, Process Total<-Total+Num, Output Total, Oval(End).[4]

4a: Input module[1], processing module (calculate grades)[1], output module[1]

b: Hiding how scores entered, focusing on grade calculation[1];
using Grade function[1]

c: Easier to understand/different programmers/easier testing[2]

5a: Each sensor represented by simple readSensor function[1], hiding hardware[1],
reduces complexity[1]

b: Data collection, storage, analysis, alerts, UI as separate modules[2]

--- ANSWER KEY 5B (35 marks) -----

1: Comp1:15!=42, Comp2:23!=42, Comp3:8!=42, Comp4:42=42 found pos3. 4 comps.[3]

2: 500[1]. Worst=target last/not present, must check every item[1]. [2]

3: Works on unsorted data / simpler[1]

4: Initial Low=0,High=9; Mid=4,List[4]=31, found! 1 comparison.[4]

5: Low=0,High=9; Mid=4(31,100>31,Low=5); Mid=7(67,100>67,Low=8);

Mid=8(89,100>89,Low=9); Mid=9(95,100>95,Low=10); Low>High, not found. 4 comps.[3]

6: $\log_2(1048576)=20$ [1]

7: Initial:[7,2,9,1,5]; P1:[2,7,1,5,9]; P2:[2,1,5,7,9]; P3:[1,2,5,7,9][5]

8: FOR i<-0 TO LENGTH-2: FOR j<-0 TO LENGTH-2-i:

IF List[j]>List[j+1] THEN Temp<-List[j]; List[j]<-List[j+1];

List[j+1]<-Temp; ENDIF; ENDFOR; ENDFOR [4]

9: n items need $\sim n^2$ comparisons (nested loops)[1]

10: |i|A|B|C| -> |1|2|3|2| |2|2|3|4| |3|2|3|6| [4]

11: |X|Output| -> |7|7| |4|4| |1|1| [3]

12: |N|Factorial|i| -> |6|6|5| |6|30|4| |6|120|3| |6|360|2| |6|720|1| |6|720|0|

Output:720 [4]

WEEK 6: PYTHON PROGRAMMING

Syllabus Ref: T8 (Programming) | 2 lessons, 2 worksheets, answer keys

--- LESSON 6.1 — CODE READING: PREDICT, DEBUG & CORRECT -----

Topic: Python Code Reading | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/2 8.1-8.5

Learning Objective: Predict output, identify/correct syntax/logic/runtime errors,
explain what code does.

Key Vocabulary: syntax error (语法错误 / yufa cuowu),

logic error (逻辑错误 / luoji cuowu), runtime error (运行时错误),
indentation (缩进 / suojin), string slicing (字符串切片 / zifuchuan qiepian),
nested loop (嵌套循环 / qiantao xunhuan), list (列表 / liebiao)

Lesson Structure (40 min):

0-3 | DO NOW: `x=5, y="5", print(x+y)`. "What happens? Why?"

(TypeError - cannot add int+str. Correct: `str(x)+y="55"` or `x+int(y)=10`)

3-12 | DIRECT INSTRUCTION:

COMMON ERRORS: Syntax (missing colons, wrong quotes, misspelled keywords, unclosed brackets). Indentation (Python uses 4 spaces, mixing tabs/spaces).

Logic (= instead of ==, off-by-one in range, integer division $5/2=2.5$ in Py3, modifying list while iterating). Runtime (index out of range, div by zero, ValueError, FileNotFoundError, TypeError).

CONSTRUCTS REVIEW: Variables (`=, +, *, /, //, %, **`), Comparison (`==, !=, <, >, <=, >=`), Logical (`and, or, not`), Selection (`if/elif/else`), Iteration (`for i in range(10)` gives 0-9, `while`), String (`text[0:5]`, `len()`, `.upper()`, `.lower()`, `.find()`), List (`myList[0]`, `.append()`, `.sort()`, `len()`, `sum()`), Nested loops (`for row in range(3): for col in range(3):`).

12-25 | GUIDED PRACTICE: Code reading in pairs. Teacher models first example:

`total=0; for i in range(1,6): if i%2==0: total=total+i; print(total)`

Answer: 6 (adds 2+4). Students then work through 4 more independently.

25-35 | INDEPENDENT PRACTICE: Worksheet 6A.

35-40 | PLENARY: Reveal answers, self-mark. HW: Complete Worksheet 6A.

Revise string slicing and lists for L6.2.

EAL Support: Line-numbered code annotations, Chinese comments allowed in working, colour-coded syntax highlighting, pair programming.

Differentiation:

Emerging: Simple code (single loops, basic IF), trace step-by-step

Developing: Predict nested structures, identify all error types, explain code

Secure: Predict complex nested code, subtle logic errors, optimise code

Resources: Slides, code examples, Worksheet 6A, highlighters.

--- LESSON 6.2 — CODE WRITING: ARRAYS, STRINGS, FILES & FUNCTIONS -----

Topic: Python Programming | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/2 8.5-8.8

Learning Objective: Write Python with lists (1D/2D), strings, file operations, define/use functions with parameters/return values.

Key Vocabulary: function (函数 / hanshu), parameter (参数 / canshu),

return value (返回值 / fanhui zhi), file handling (文件处理 / wenjian chuli),
CSV (逗号分隔值 / douthao fenge zhi), append (添加 / tianjia)

Lesson Structure (40 min):

0-3 | DO NOW: "Write function greet(name) that prints 'Hello [name]!'"

3-12 | DIRECT INSTRUCTION:

FUNCTIONS: def name(params): body; return value. Key rules: colon,
indented body, return sends back value.

1D LISTS: scores=[]; for i in range(5): mark=int(input()); scores.append(mark)

Finding highest: highest=scores[0]; for s in scores: if s>highest: highest=s

Linear search: def find(my_list,target): for i in range(len(my_list)):
if my_list[i]==target: return i; return -1

2D LISTS: grid=[[1,2,3],[4,5,6],[7,8,9]]; grid[0][0]=1; grid[1][2]=6

Nested iteration: for row in range(3): for col in range(3):

print(grid[row][col],end=" ")

STRING HANDLING: text[7:12], text.split(", "), ".join(["a","b","c"])

Validation: check_password(password): if len(password)<8: return False

FILE HANDLING: open("file","r") for read, "w" for write.

for line in file: data=line.strip().split(",")

CSV: import csv; csv.reader(file); csv.writer(file)

With statement: with open("file") as f: (auto-closes)

12-25 | GUIDED PRACTICE: "Write function count_even(numbers) returning count
of even numbers." Then: "Write function that takes filename and returns
number of lines." Pairs.

25-35 | INDEPENDENT PRACTICE: Worksheet 6B.

35-40 | PLENARY: Review guided practice answers. HW: Complete Worksheet 6B.

EAL Support: Code templates with bilingual comments, fill-in-blank function defs,
visual memory diagrams for lists, bilingual Python glossary.

Differentiation:

Emerging: Fill-in-blank code, single-construct problems, templates

Developing: Complete functions, 1D lists, file operations, combine constructs

Secure: 2D list operations, file input->process->output, efficient solutions

Resources: Slides, Python IDE, Worksheet 6B, sample data files.

--- WORKSHEET 6A — CODE READING: PREDICT, DEBUG & CORRECT -----

Total: 40 marks | Name: _____ Class: _____ Date: _____

SECTION A: PREDICT OUTPUT (16 marks)

For each, write EXACT output.

```

72.x=10; y=3; print(x//y); print(x%y); print(x/y) [3]
73.total=0; for i in range(5): total=total+i*2; print(total) [2]
74.name="COMPUTER"; print(name[2:5]); print(name[-1]); print(len(name)) [3]
75.nums=[5,2,8,1,9]; nums.sort(); print(nums[0]); print(nums[-1]); print(len(nums)) [3]
76.def mystery(n): if n<=1: return n; return mystery(n-1)+mystery(n-2); print(mystery(6))
    [2]
77.count=0; for i in range(3): for j in range(3): if i==j: count=count+1; print(count) [2]
78.text="hello world"; result=""; for c in text: if c!=" ": result=result+c.upper(); print(result)
    [1]

```

SECTION B: IDENTIFY AND CORRECT ERRORS (12 marks)

Each has one or more errors. Identify and write corrected version.

```

79.age=input("Age: "); if age>18: print("Adult") else: print("Child") [2]
80.for i in range(10): print(i); i=i+1 [4]
81.numbers=[10,20,30,40,50]; print(numbers[5]) [2]
82.def add(a,b): result=a+b; print(add(3,4)) [4]

```

SECTION C: EXPLAIN THE CODE (12 marks)

12. Explain what this does (to another student):

```

scores=[78,85,62,90,55,88]; total=0; count=0; for s in scores:
    if s>=60: total=total+s; count=count+1; average=total/count;
    print("Average pass mark:",average) [4]

```

13. Explain purpose and trace for n=4:

```

def factorial(n): result=1; for i in range(1,n+1): result=result*i; return result
print(factorial(4)) [4]

```

14. Explain and predict output:

```

data=[3,1,4,1,5,9,2,6]; unique=[]; for n in data: if n not in unique:
    unique.append(n); print(unique); print(len(unique)) [4]

```

--- WORKSHEET 6B — CODE WRITING: FUNCTIONS, ARRAYS, FILES -----

Total: 40 marks | Name: _____ Class: _____ Date: _____

SECTION A: FUNCTIONS (12 marks)

1. Write `max_of_three(a,b,c)` returning largest. [4]

2. Write `is_even(n)` returning True/False. [3]

3. Write `reverse_string(text)` returning reversed. [3]

b) Call with "COMPUTER", print result. [2]

SECTION B: LIST OPERATIONS (10 marks)

83. Write program: a) Input 5 names into list [2] b) Output alphabetically [2]
c) Count names starting with "A" [3]

84. Create 3x3 game board (zeros). Input row+col, place 1. Print board. [3]

SECTION C: FILE HANDLING (10 marks)

85. "scores.txt" has one number per line. Read all, calculate and print
total and average. [6]

86. Open "output.txt" for writing. Write 1-20, one per line. Close file. [4]

SECTION D: PRACTICAL CHALLENGES (8 marks)

87. Password checker: >=8 chars, at least one digit. Output Valid/Invalid. [4]

88. "students.csv": Name,Math,English,Science. Read and print each name
with average across three subjects. [4]

--- ANSWER KEY 6A (40 marks) -----

1: 3[1], 1[1], 3.333...[1]

2: 20[2]

3: MPU[1], R[1], 8[1]

4: 1[1], 9[1], 5[1]

5: 8[2]

6: 3[2]

7: HELLOWORLD[1]

8: Error: input returns string[1]. Correct: age=int(input(...)) [1]

9: Missing colons[2]. Correct: for i in range(10): print(i) [2]

10: Error: index 5 out of range (0-4)[1]. Correct: numbers[4] or any 0-4[1]

11: Missing colon[1], no return[1], name misspelled[1].

Correct: def add(a,b): return a+b; print(add(3,4)) [4]

12: Calculates average of passing scores (>=60)[1]. Initialises total/count[1].

Loops checking each score[1]. Calculates average and prints[1]. [4]

13: Calculates factorial (n!)[1]. Trace: i=1:1, i=2:2, i=3:6, i=4:24[2].

Output:24[1]. [4]

14: Removes duplicates, keeps first occurrence[1]. Creates empty list[1].

Checks not in list before appending[1]. Output:[3,1,4,5,9,2,6], 7[1].[4]

--- ANSWER KEY 6B (40 marks) -----

1: def max_of_three(a,b,c): if a>=b and a>=c: return a; elif b>=c: return b;
else: return c [4]

2: def is_even(n): return n%2==0 [3]

3: def reverse_string(t): return t[::-1] [3]

```

b: print(reverse_string("COMPUTER")) -> RETUPMOC [2]
4a: names=[]; for i in range(5): n=input(); names.append(n) [2]
    b: names.sort(); for n in names: print(n) [2]
    c: count names starting with A using loop and condition [3]
5: board=[[0,0,0],[0,0,0],[0,0,0]]; row=int(input()); col=int(input());
    board[row][col]=1; print with nested loop [3]
6: file=open("scores.txt"); total=0; count=0;
    for line in file: num=int(line.strip()); total+=num; count+=1;
    file.close(); print(total, total/count) [6]
7: file=open("output.txt","w"); for i in range(1,21): file.write(str(i)+"\n");
    file.close() [4]
8: password=input(); valid=len(password)>=8; has_digit=False;
    for c in password: if c.isdigit(): has_digit=True;
    if valid and has_digit: print("Valid") else: print("Invalid") [4]
9: file=open("students.csv"); for line in file: data=line.strip().split(",");
    name=data[0]; avg=(int(data[1])+int(data[2])+int(data[3]))/3;
    print(name,avg); file.close() [4]

```

WEEK 7: DATABASES, BOOLEAN LOGIC & VALIDATION

Syllabus Ref: T9 (Databases), T10 (Boolean logic) | 2 lessons, 2 worksheets

--- LESSON 7.1 — SQL & DATABASE DESIGN -----

Topic: Databases | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/2 9.1-9.6

Learning Objective: Write SQL (SELECT,WHERE,ORDER BY,LIKE,BETWEEN,IN), explain flat-file vs relational, identify keys, describe normalisation (1-3NF), draw ERDs.

Key Vocabulary: SQL (结构化查询语言 / jie gou hua chaxun yuyan), database (数据库 / shu ju ku), table (表 / biao), record/row (记录 / ji lu), field/column (字段 / zi duan), primary key (主键 / zhu jian), foreign key (外键 / wai jian), relationship (关系 / guan xi), normalisation (规范化 / gui fan hua), flat-file (平面文件 / ping mian wen jian), relational (关系 / guan xi), ERD (实体关系图 / shi ti guan xi tu), wildcard (通配符 / tong pei fu)

Lesson Structure (40 min):

0-3 | DO NOW: "Write SQL to find all students in class 10A with scores >80."

3-12 | DIRECT INSTRUCTION:

SQL COMMANDS: SELECT Name,Score FROM Students; SELECT * FROM Students;

SELECT DISTINCT Class FROM Students;

WHERE: SELECT * FROM Students WHERE Score>80;

AND/OR: WHERE Age>=15 AND Age<=16; WHERE Class='10A' OR Class='10B'

ORDER BY: ORDER BY Score DESC; ORDER BY Age,Name

LIKE: WHERE Name LIKE 'A%' (starts A), '%e%' (contains e), '___' (3 chars)

BETWEEN: WHERE Score BETWEEN 70 AND 85 (=Score>=70 AND Score<=85)

IN: WHERE Class IN ('10A','10B','10C')

AGGREGATES: COUNT(*), AVG(Score), MAX(Score), MIN(Score), SUM(Score)

GROUP BY: SELECT Class,AVG(Score) FROM Students GROUP BY Class

DATABASE DESIGN: Primary key=unique identifier (cannot be NULL).

Foreign key=field referencing primary key of another table.

Flat-file=single table (simple, causes redundancy).

Relational=multiple linked tables (no redundancy, easier maintenance).

NORMALISATION: 1NF=all fields atomic, no repeating groups.

2NF=in 1NF+no partial dependencies. 3NF=in 2NF+no transitive dependencies.

ERD: One-to-many: [Student]----<[Enrollment]. Many-to-many resolved through junction table.

12-25| GUIDED PRACTICE: Given Students(StudentID,Name,Age,ClassID),
Classes(ClassID,ClassName,Teacher), Grades(GradeID,StudentID,Subject,Score).
Write: 1) Students aged 16+ 2) Teacher of 10A 3) Grades for S001
4) Average Math score.

25-35| INDEPENDENT PRACTICE: Worksheet 7A.

35-40| PLENARY: "SQL: find names starting with A, score>75, sorted desc."

HW: Worksheet 7A. Revise Boolean gates for L7.2.

EAL Support: SQL reference card (bilingual), database diagrams with Chinese labels, mnemonic "SELECT what FROM where WHERE condition ORDERed how", sentence frames for SQL.

Differentiation:

Emerging: Basic SELECT/WHERE, identify primary keys, flat-file vs relational

Developing: Complex queries (AND/OR,ORDER BY,LIKE), explain 1NF/2NF, simple ERDs, identify foreign keys

Secure: All query types, normalise to 3NF with justification, complex ERDs with cardinality, explain why normalisation reduces redundancy

Resources: Slides, SQL reference cards, schema diagram, Worksheet 7A.

--- LESSON 7.2 — BOOLEAN LOGIC & VALIDATION -----

Topic: Boolean Logic & Data Validation | Duration: 40 min | Lesson Type: Revision

Syllabus Ref: 0478/2 10.1-10.4

Learning Objective: Complete truth tables for all 6 gates (AND,OR,NOT,NAND,NOR,XOR), write/simplify Boolean expressions, draw circuits, write Python validation (range,type,length,presence,format).

Key Vocabulary: Boolean logic (布尔逻辑 / Bu'er luoji),
logic gate (逻辑门 / luoji men), truth table (真值表 / zhenzhi biao),
Boolean expression (布尔表达式), validation (验证 / yanzheng),
range check (范围检查 / fanwei jiancha), type check (类型检查),
length check (长度检查 / changdu jiancha), presence check (存在检查),
format check (格式检查 / geshi jiancha)

Lesson Structure (40 min):

0-3 | DO NOW: "Complete: A B AAND B AOR B; 00?_? 01?_? 10?_? 11?_?"

3-12 | DIRECT INSTRUCTION:

ALL 6 GATES:

AND: output=1 only if ALL inputs=1. Analogy: both switches ON.

OR: output=1 if ANY input=1. Analogy: either switch ON.

NOT: inverts input. Analogy: opposite switch.

NAND: AND+NOT. Output=0 only if ALL=1. Universal gate.

NOR: OR+NOT. Output=1 only if ALL=0. Universal gate.

XOR: output=1 if inputs DIFFERENT. Analogy: exactly one switch ON.

3-INPUT: For AND with 3 inputs, output=1 only when A=B=C=1. $2^3=8$ rows.

CIRCUITS: $Q=(A \text{ AND } B) \text{ OR } (\text{NOT } C) \rightarrow$ AND gate (A,B) $\rightarrow$ OR gate;

NOT(C) $\rightarrow$ OR gate.

VALIDATION PYTHON:

Range: while score<0 or score>100: (re-prompt)

Type: while not user_input.isdigit(): (re-prompt)

Length: while len(password)<6 or len(password)>12: (re-prompt)

Presence: while name=="": (re-prompt)

Format: while "@" not in email or "." not in email: (re-prompt)

12-25 | GUIDED PRACTICE: Circuit for $Q=(A \text{ OR } B) \text{ AND } (B \text{ NAND } C)$ - complete truth table. Validation: UK postcode format check.

25-35 | INDEPENDENT PRACTICE: Worksheet 7B.

35-40 | PLENARY: Quick-fire gate quiz. HW: Worksheet 7B.

EAL Support: Gate symbol reference with icons, pre-printed truth table templates, colour-code (AND=blue, OR=green, NOT=red), physical gate cards.

Differentiation:

Emerging: 2-input tables with guidance, basic range/presence checks

Developing: All 2-input tables, circuits from expressions, all 5 validation types

Secure: 3-input tables, simplify expressions, complex circuits, combined validation, explain why NAND is universal gate

Resources: Slides, gate cards, truth table templates, Worksheet 7B, Python IDE.

--- WORKSHEET 7A — SQL QUERIES & DATABASE DESIGN -----

Total: 40 marks | Name: _____ Class: ____ Date: _____

Use this schema: Students(StudentID[PKey],FirstName,LastName,Age,ClassID[FKey])

Classes(ClassID[PKey],ClassName,Room,Teacher)

Enrollments(EnrollmentID[PKey],StudentID[FKey],Subject,Score)

SECTION A: SQL QUERIES (20 marks)

- 1. List all students' first and last names. [1]**
- 2. Find all students aged 16. [2]**
- 3. List classes taught by "Mr. Smith". [2]**
- 4. Find enrollments with score ≥ 80 . [2]**
- 5. Students in "10A" or "10B", sorted by last name. [3]**
- 6. Students whose last name starts with "W". [2]**
- 7. Enrollments with scores between 60 and 80. [2]**
- 8. Count students in each class. [2]**
- 9. Average score for "Mathematics". [2]**

SECTION B: DATABASE DESIGN (12 marks)

10. Flat-file vs relational. Advantage of relational. [3]

89. a) One primary key b) One foreign key c) Purpose of foreign key [4]

90. Table: StudentID,Name,Subject1,Subject2,Subject3. Why not 1NF?

How normalise? [3]

91. ERD: Students, Classes, Enrollments. Show cardinality. [2]

SECTION C: ADVANCED SQL (8 marks)

92. Students who scored >85 (first name, subject, score). [3]

15. Highest score in each subject. [2]

16. Count students who scored <50. [3]

--- WORKSHEET 7B — BOOLEAN LOGIC & VALIDATION -----

Total: 35 marks | Name: _____ Class: ____ Date: _____

SECTION A: TRUTH TABLES (12 marks)

1. Complete 2-input NAND: |A|B|A NAND B| (all 4 rows) [2]

2. Complete 2-input XOR: |A|B|A XOR B| (all 4 rows) [2]

93. Complete: $Q = (A \text{ AND } B) \text{ OR } (\text{NOT } C)$ (all 8 rows, Q only needs 4 marks) [4]

4. Complete 3-input AND (all 8 rows) [4]

SECTION B: LOGIC CIRCUITS (8 marks)

5. Draw circuit for $Q = (A \text{ OR } B) \text{ AND } C$ [3]

94. Write expression: "Q=1 when A=0 AND B=1, OR when C=1" [2]

7. Simplify: $(A \text{ AND } B) \text{ OR } (A \text{ AND NOT } B)$. Explain. [3]

SECTION C: VALIDATION CODE (15 marks)

8. Python: range check 0-100, keep asking until valid. [3]

9. Python: length check username 4-15 characters. [3]

95. Python: format check phone starts with "1", exactly 11 digits. [3]

11. Python: type check positive integer, keep asking. [3]

96. Function validate_password: >=8 chars, at least one digit. Returns T/F. [3]

--- ANSWER KEY 7A (40 marks) -----

1: SELECT FirstName, LastName FROM Students; [1]

2: SELECT * FROM Students WHERE Age=16; [2]

3: SELECT * FROM Classes WHERE Teacher='Mr. Smith'; [2]

4: SELECT * FROM Enrollments WHERE Score>=80; [2]

5: SELECT * FROM Students WHERE ClassName='10A' OR ClassName='10B'
ORDER BY LastName; [3]

6: SELECT * FROM Students WHERE LastName LIKE 'W%'; [2]

7: SELECT * FROM Enrollments WHERE Score BETWEEN 60 AND 80; [2]

8: SELECT ClassID,COUNT(*) FROM Students GROUP BY ClassID; [2]

9: SELECT AVG(Score) FROM Enrollments WHERE Subject='Mathematics'; [2]

10: Flat-file=single table[1], relational=multiple linked tables[1].
Advantage: reduced redundancy/easier maintenance[1]. [3]

11a: StudentID/ClassID/EnrollmentID[1] b: ClassID in Students/StudentID in Enroll[1]
c: Links tables[1], references primary key of another table[1]. [4]

12: Repeating groups (Subject1,2,3)[1]. Separate table: StudentID,Subject[1],
any number of subjects[1]. [3]

13: Students 1----* Enrollments, Classes 1----* Students [2]

14: SELECT S.FirstName,E.Subject,E.Score FROM Students S
INNER JOIN Enrollments E ON S.StudentID=E.StudentID WHERE E.Score>85; [3]

15: SELECT Subject,MAX(Score) FROM Enrollments GROUP BY Subject; [2]

16: SELECT COUNT(*) FROM Enrollments WHERE Score<50; [3]

--- ANSWER KEY 7B (35 marks) -----

1: NAND: 00->1, 01->1, 10->1, 11->0 [2]

2: XOR: 00->0, 01->1, 10->1, 11->0 [2]

3: (A AND B) OR (NOT C): 000->1, 001->0, 010->1, 011->0, 100->1, 101->0,
110->1, 111->1 [4]

4: 3-input AND: 000->0, 001->0, 010->0, 011->0, 100->0, 101->0, 110->0, 111->1[4]

5: OR gate(A,B) -> AND gate(OR_output, C) -> Q [3]

6: Q=(NOT A AND B) OR C [2]

7: Simplified=A[2]. (A AND B) covers A=1,B=1; (A AND NOT B) covers A=1,B=0.
Together all A=1 cases regardless of B[1]. [3]

8: score=int(input()); while score<0 or score>100:
score=int(input("Enter 0-100: ")) [3]

9: username=input(); while len(username)<4 or len(username)>15: (re-prompt)[3]

10: phone=input(); while len(phone)!=11 or phone[0]!="1" or not phone.isdigit():
(re-prompt) [3]

11: ui=input(); while not ui.isdigit() or int(ui)<=0: (re-prompt)[3]

12: def validate_password(p): return len(p)>=8 and any(c.isdigit() for c in p)[3]
(Accept loop-based solution)

WEEK 8: PAPER 2 TIMED PRACTICE + 15-MARK SCENARIO

Syllabus Ref: T7-10 (Complete Paper 2) | 2 lessons, timed paper + scenario

--- LESSON 8.1 — THE 15-MARK SCENARIO: STRATEGY & PRACTICE -----

Topic: 15-Mark Scenario Strategy | Duration: 40 min | Lesson Type: Strategy

Syllabus Ref: 0478/2 7.1, 8.1, 8.2

Learning Objective: Apply systematic 5-step approach to 15-mark scenario, plan with pseudocode before Python, write structured code with comments.

Key Vocabulary: scenario question (情景题 / qingjing ti),

plan first (先计划 / xian jihua), comments (注释 / zhushi),

structure (结构 / jiegou)

Lesson Structure (40 min):

0-3 | DO NOW: "15-mark scenario = 20% of Paper 2. What grade could you get with perfect 15 marks and nothing else?" ($15/75=20\%$. At C boundary ~44, so 15 marks + 29/60 remaining = 48% of rest. Very achievable C/A.)

3-15 | DIRECT INSTRUCTION: THE 5-STEP STRATEGY

STEP 1 READ (3 min): Read twice. Underline INPUTS, OUTPUTS, PROCESSES.

Identify data structures. Note constraints.

STEP 2 PLAN (7 min): Write pseudocode FIRST. Break into sub-tasks:

// Step 1: Read data // Step 2: Validate // Step 3: Process

// Step 4: Output. Pseudocode earns marks even with Python errors!

STEP 3 CODE (15 min): Convert pseudocode line-by-line. Meaningful names.

Comments. Input validation. Edge cases.

STEP 4 TEST (3 min): Trace with sample data. First record? Last record?

Invalid input?

STEP 5 CHECK (2 min): All sections commented. Consistent indentation.

Clear variable names.

MARK TYPICAL: Planning 2-3, Input 2-3, Processing 4-6, Output 2-3,

Validation/comments 2-3.

COMMON MISTAKES: Writing Python immediately (no plan), no comments,

no validation, using variables before defining, not closing files,

off-by-one errors, not testing with sample data.

15-25 | GUIDED PRACTICE: WORKED EXAMPLE

SCENARIO: "scores.csv has StudentName,Math,English,Science per line.

Read file, calculate each student's average, write to averages.txt

as 'Name: Average'. Only include averages ≥ 50 ."

Teacher models all 5 steps with full pseudocode and Python code.

25-35| INDEPENDENT PRACTICE: Students receive 8-mark scenario, apply 5-step.

Teacher circulates, checks planning before coding.

35-40| PLENARY: "MOST important thing about 15-mark question?"

(Plan first with pseudocode!) HW: Complete 8-mark scenario.

EAL Support: 5-step poster (bilingual), sentence starters for comments, scaffolded planning template, visual flowchart.

Differentiation:

Emerging: Planning template with teacher support, basic pseudocode, simple Python (file open, basic loop)

Developing: All 5 steps independently, full pseudocode+Python, basic validation and comments

Secure: All steps efficiently, optimal Python with comprehensive validation, error handling, detailed comments, critique given solutions

Resources: 5-step poster (A3), planning template, practice scenario, CSV data.

--- LESSON 8.2 — TIMED PRACTICE: PAPER 2 STYLE -----

Topic: Paper 2 Mixed | Duration: 40 min | Lesson Type: Exam Practice

Syllabus Ref: 0478/2 T7-10

Learning Objective: Practice 45-minute Paper 2 section under timed conditions.

Lesson Structure (40 min):

0-3 | DO NOW: "Paper 2: 1h45m for 75 marks. 15-mark scenario: 25-30 min.

ALWAYS pseudocode before Python. Show ALL working for trace tables.

Write full SQL queries."

3-35 | TIMED PRACTICE: Students complete Timed Practice Paper 2. Invigilate.

10-min warning at 25 min.

35-40| COLLECTION: "Review next lesson. How was timing? Which was hardest?"

EAL Support: Extra 5 min. Vocabulary list on request.

Differentiation:

Emerging: Complete Section A (trace tables + pseudocode), attempt B (SQL+Boolean)

Developing: Complete all except scenario, attempt scenario with pseudocode plan

Secure: Complete entire paper including scenario with pseudocode+Python+comments

--- PRACTICE SCENARIO (8 MARKS) — FOR LESSON 8.1 -----

Total: 8 marks | Name: _____ Class: ____ Date: _____

SCENARIO: "steps.txt" has Username,Steps per line (e.g., "Alice,8500").

Write Python that: 1) Reads all data 2) Counts users with ≥ 10000 steps

3) Calculates average steps across ALL users 4) Outputs count and average.

PLANNING SPACE:

Step 1 - Inputs/Processes/Outputs: _____

Step 2 - Pseudocode: _____

Step 3 - Python code:

```
```python
```

```
```
```

--- MARK SCENARIO 8 marks -----

Planning/Pseudocode [2] File handling [2] Processing [3] Output [1]

--- TIMED PRACTICE PAPER 2 (40 MINUTES, 28 MARKS) -----

Name: _____ Class: _____ Date: _____ Time: 40 min Total: 28

Instructions: Answer ALL. Show all working. Use Cambridge pseudocode.

Write clear Python with comments.

SECTION A: ALGORITHMS & TRACE TABLES (8 marks)

97. Trace: Start=1, End=5; Total=0; i=Start; WHILE i<=End: Total=Total+i*i;
i=i+1; OUTPUT Total. Complete trace table. [4]

98. Write pseudocode: find largest in 10 numbers input by user. [4]

SECTION B: SQL & BOOLEAN LOGIC (8 marks)

99. Table Players(PlayerID, Name, Team, Score). SQL: a) All in "Red" [1]
b) Highest score [1] c) Scores 50-100 sorted descending [2]

4. Complete: $Q = (A \text{ AND NOT } B) \text{ OR } C$ (all 8 rows) [4]

SECTION C: PYTHON PROGRAMMING (12 marks)

5. Code has errors. Identify and correct:

```
def calculate_average(numbers): total=0; for num in numbers:
```

```
total=total+num; average=total/len(numbers); print(average)
```

```
scores=[85,90,78,92]; result=calculate_averag(scores) [4]
```

100. Write code: a) Input 5 names into list [2] b) Output in reverse [2]

101. SCENARIO (8 marks): "sales.txt" has ProductName, Quantity, PricePerUnit.

Read all sales, calculate total revenue (Quantity*Price each line),

output total revenue. Include comments.

--- MARK SCHEME PAPER 2 (28 marks) -----

1: |i|i*i|Total|i<=End?| 1|1|1|T 2|4|5|T 3|9|14|T 4|16|30|T 5|25|55|T
6|36|91|F| Output:55 [4]

2: Input 10 numbers into array[1], Largest=first[1], loop comparing[1],
output Largest[1]. [4]

3a: SELECT * FROM Players WHERE Team='Red'; [1]
 b: SELECT MAX(Score) FROM Players; [1]
 c: SELECT * FROM Players WHERE Score BETWEEN 50 AND 100 ORDER BY Score DESC; [2]
 4: 000->1, 001->0, 010->1, 011->0, 100->1, 101->1, 110->0, 111->1 [4]
 5: Missing colons[1], no return[1], name misspelled[1], should print result[1].
 Correct: def calculate_average(numbers): return sum(numbers)/len(numbers) [4]
 6a: names=[]; for i in range(5): names.append(input()) [2]
 b: for n in reversed(names): print(n) [2]
 7: File open/read[2], split and convert[2], calculate revenue[2], output[1],
 comments[1]. [8]

--- GAP ANALYSIS TEMPLATE — PAPER 2 TIMED PRACTICE -----

Name: _____ Date: _____ Score: ____/28

| Topic | Q | Avail | Score | % | Priority |
|---------------|---|-------|-------|---|----------|
| Trace Tables | 1 | 4 | | | |
| Pseudocode | 2 | 4 | | | |
| SQL | 3 | 4 | | | |
| Boolean Logic | 4 | 4 | | | |
| Code Debug | 5 | 4 | | | |
| Code Writing | 6 | 4 | | | |
| Scenario | 7 | 8 | | | |

TOP 3 PRIORITIES: 1: _____ 2: _____ 3: _____

UNIT 3: MOCK SERIES 1

#

WEEK 9: MOCK EXAM 2 — PAPER 1

Syllabus Ref: T1-6 (Complete Paper 1) | Full 75-mark mock + mark scheme

--- LESSON 9.1 — MOCK EXAM: PAPER 1 (FULL 1H 45MIN) -----

Topic: Full Paper 1 Mock Examination | Duration: 40 min lesson block

Lesson Type: Formal Examination | Syllabus Ref: 0478/1 All T1-6

Learning Objective: Complete full Paper 1 mock under exam conditions.

Exam Conditions:

- 1 hour 45 minutes (105 min) for 75 marks
- Silent working, no phones/smart watches, no communication
- Calculators NOT permitted
- Answers in pen (pencil only for diagrams)

Materials: Question paper, answer booklet, rough paper (collected, not marked)

Invigilation:

- Seating: single desks, 1 metre apart
- Distribute face-down, "You may begin" when all have papers
- Time announcements: 60 min, 30 min, 15 min, 5 min remaining
- No hints or clarifications
- Record any incidents

--- LESSON 9.2 — REVIEW, MARK SCHEME & GAP ANALYSIS -----

Topic: Paper 1 Mock Review | Duration: 40 min | Lesson Type: Review

Learning Objective: Self-mark mock, identify gaps, create targeted revision plan.

Lesson Structure (40 min):

0-3 | DO NOW: "Before seeing marks, predict your score /75. Write it down."

3-25 | MARK SCHEME WALKTHROUGH: Project mark scheme. Each Q: read, show marks,

"Who got full?", common errors. Key errors: padding zeros, explain vs

define, compare one-sided only, forgetting /8 or /1024, MAR/MDR confusion.

25-32| SELF-MARKING: Green pen. Total at top. Teacher circulates.

32-38| INDIVIDUAL GAP ANALYSIS: Topic breakdown with scores and priorities.

38-40| CLASS ANALYSIS: Class average, highest, common errors, revision focus.

HW: Revise weakest topic using notes and worksheets W1-3.

EAL Support: Bilingual gap analysis, individual target discussion,
extra support materials for weak topics.

Differentiation:

Emerging: Teacher-assisted marking, simplified analysis, specific worksheets

Developing: Self-mark, full gap analysis, own revision targets

Secure: Self-mark, full analysis, create resources for peers, lead groups

MOCK EXAM 2 — PAPER 1

CIE IGCSE COMPUTER SCIENCE (0478/11)

FULL MOCK EXAMINATION

Time: 1 hour 45 minutes | Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

Write your name, class, and date. Answer ALL questions.

Show ALL working for calculations. No calculators.

QUESTION 1: DATA REPRESENTATION (10 marks)

(a) Convert denary 219 to 8-bit binary. [1]

(b) Convert binary 1010 1100 to denary. Show working. [2]

(c) Convert hex 3F to binary. [1]

(d) Explain one reason why hexadecimal is used instead of binary. [2]

(e) Image: 800x600 pixels, 24-bit colour depth.

(i) Calculate file size in KB. Show working. [2]

(ii) Saved with lossy compression to 150 KB. One advantage and one disadvantage of lossy for this image. [2]

QUESTION 2: TEXT, SOUND & COMPRESSION (8 marks)

(a) ASCII uses 7 bits per character.

(i) How many characters can ASCII represent? Show working. [1]

(ii) Explain why ASCII is not suitable for Chinese text. [2]

(b) Sound: sampling rate 44,100 Hz, bit depth 16 bits.

(i) Explain sampling rate and bit depth. [2]

(ii) Recorded for 2 minutes in stereo. Calculate file size in MB.

(1 MB = 1,048,576 bytes) Show working. [3]

QUESTION 3: CPU ARCHITECTURE (10 marks)

(a) Draw labelled Von Neumann architecture showing CPU (ALU, CU, registers), Memory, I/O devices, Buses. [4]

(b) State purpose: (i) PC [1] (ii) MAR [1] (iii) CIR [1]

(c) Describe role of ALU. [1]

(d) Program stored at address 100: 100=LDA 150, 101=ADD 151, 102=STO 152, 103=HLT. Mem[150]=25, Mem[151]=17.

Complete FDE trace table (PC/MAR/MDR/CIR/ACC):

| Cycle | PC | MAR | MDR | CIR | ACC | Description |
|-------|----|-----|-----|-----|-----|-------------|
|-------|----|-----|-----|-----|-----|-------------|

| | | | | | | |
|---|-----|-----|---------|---------|--|-------|
| 1 | 100 | 100 | LDA 150 | LDA 150 | | Fetch |
|---|-----|-----|---------|---------|--|-------|

[2]

QUESTION 4: MEMORY & STORAGE (8 marks)

(a) Order by access speed (fastest to slowest): Cache, RAM, Registers, Virtual Memory [2]

(b) Explain purpose of cache memory. [2]

(c) Compare magnetic, optical, solid-state. For each: one example, one advantage.

Magnetic: Example_____ Advantage_____

Optical: Example_____ Advantage_____

Solid-state: Example_____ Advantage_____

[6]

QUESTION 5: NETWORKS & THE INTERNET (10 marks)

(a) Explain Internet vs WWW. [2]

(b) Describe process when user types URL into browser, including DNS role.[4]

(c) Explain difference between router and switch. [2]

(d) State two advantages of IPv6 over IPv4. [2]

QUESTION 6: SECURITY & EMERGING TECH (10 marks)

(a) Describe: (i) Virus [1] (ii) Worm [1] (iii) Trojan [1]

(b) Describe three protection measures against malware/security threats.[6]

(c) Automated greenhouse uses sensors and actuators.

(i) Name one sensor and one actuator. [2]

Sensor:_____ Actuator:_____

(ii) Explain one advantage of automation in this context. [1]

QUESTION 7: SOFTWARE & OPERATING SYSTEMS (8 marks)

(a) Describe three functions of an operating system. [6]

(b) Explain what happens when an interrupt occurs. [2]

QUESTION 8: DATA TRANSMISSION & ENCRYPTION (11 marks)

(a) Explain packet switching (sender, network, receiver). [4]

(b) Even parity: 1101 011_

(i) Calculate parity bit. Show working. [1]

(ii) Explain how parity detects errors. [2]

(c) Explain symmetric vs asymmetric encryption. [4]

END OF PAPER

MARK SCHEME — MOCK EXAM 2 PAPER 1 (75 marks)

GRADE BOUNDARIES: A*:68-75 A:60-67 B:52-59 C:44-51 D:36-43 E:28-35 F:20-27 G:<20

Q1: DATA REPRESENTATION (10 marks)

(a) 11011011 [1]

(b) $128+32+8+4=172$ [2] (working[1], answer[1])

(c) $3=0011$, $F=1111 \rightarrow 00111111$ [1]

(d) Any two: shorter/easier to read than binary[1]; one hex digit=4 bits[1];
used for memory addresses/colour codes/MAC[1]. (max[2])

(e)(i) $800 \times 600 \times 24 = 11520000$ bits[0.5]; $/8 = 1440000$ bytes[0.5]; $/1024 = 1406.25\text{KB}$ [1]

(ii) Advantage: smaller/faster to send[1]; Disadvantage: permanent data
loss/quality reduced[1]

Q2: TEXT, SOUND & COMPRESSION (8 marks)

(a)(i) $2^7=128$ [1]

(ii) Only 128 characters[1]; Chinese has thousands, far more than ASCII[1]

(b)(i) Sampling rate: samples per second (Hz)[1]; Bit depth: bits per sample[1]

(ii) $44100 \times 16 \times 2 \times 120 = 169344000$ bits[1]; $/8 = 21168000$ bytes[1];

/1048576=20.19MB (accept 20.2)[1]

Q3: CPU ARCHITECTURE (10 marks)

- (a) CPU box with ALU[0.5], CU[0.5], registers[0.5]; Memory[0.5];
I/O[0.5]; Buses[0.5]; Labels[1]. [4]
- (b)(i) PC=address of next instruction[1] (ii) MAR=memory address accessed[1]
(iii) CIR=current instruction[1]
- (c) ALU: arithmetic and logical operations[1]
- (d) Row2: PC101,MAR150,MDR25,CIR"LDA150",ACC25
Row3: PC101,MAR101,MDR"ADD151",CIR"ADD151",ACC25
Row4: PC102,MAR151,MDR17,CIR"ADD151",ACC42
Row5: PC102,MAR102,MDR"STO152",CIR"STO152",ACC42
Row6: PC103,MAR152,MDR42,CIR"STO152",ACC42 [2]

Q4: MEMORY & STORAGE (8 marks)

- (a) Registers, Cache, RAM, Virtual Memory [2] (deduct [1] per error)
- (b) Stores frequently used data near CPU[1]; reduces waiting time[1]
- (c) Magnetic: HDD/tape[0.5]; large capacity/low cost[1.5]
Optical: CD/DVD/BD[0.5]; removable/cheap[1.5]
Solid-state: SSD/USB[0.5]; fast/no moving parts[1.5]

Q5: NETWORKS & INTERNET (10 marks)

- (a) Internet=global network of networks[1]; WWW=service on Internet using
HTTP/HTTPS[1]
- (b) Browser checks cache[0.5]; query DNS[0.5]; root->TLD->authoritative[0.5];
returns IP[0.5]; browser connects[0.5]; IP cached[0.5]. (max 4, award
best 8 half-marks)
- (c) Router=connects networks, forwards by IP/routing tables[1];
Switch=same LAN, uses MAC addresses[1]
- (d) Any two: 128-bit vs 32-bit[1]; built-in security[1]; efficient routing[1];
eliminates NAT[1]. [2]

Q6: SECURITY & EMERGING TECH (10 marks)

- (a)(i) Virus: attaches to programs, needs user action, corrupts data[1]
(ii) Worm: self-replicating, spreads across networks[1]
(iii) Trojan: disguised as legitimate, creates backdoor[1]
- (b) Any three with descriptions (max[6]): Firewall[2], Antivirus[2],
Encryption[2], Backup[2], Patches[2], 2FA[2], User Access Levels[2]
- (c)(i) Sensor: temperature/light/moisture[1]; Actuator: heater/fan/motor[1]
(ii) Consistent conditions/24-7/faster/safer[1]

Q7: SOFTWARE & OS (8 marks)

(a) Any three with descriptions (max[6]): Memory management[2], Processor scheduling[2], Device management[2], File management[2], Security[2], User interface[2]

(b) Save state[0.5]; run ISR[0.5]; restore and resume[0.5]; respond to urgent events[0.5]. [2]

Q8: TRANSMISSION & ENCRYPTION (11 marks)

(a) Split into packets[1]; header with addresses[1]; travel independently[0.5]; routers forward[0.5]; different routes[0.5]; reassembled at dest[0.5]. [4]

(b)(i) Five 1s (odd), parity=1 [1]

(ii) Sender calculates parity[0.5]; sends with data[0.5]; receiver recalculates[0.5]; mismatch=error detected[0.5]. [2]

(c) Symmetric: same key encrypt/decrypt[1]; key distribution problem[1].

Asymmetric: public encrypt, private decrypt[1]; public shared openly[1]. [4]

GAP ANALYSIS MATERIALS

INDIVIDUAL GAP ANALYSIS — MOCK EXAM 2 PAPER 1

Name: _____ Date: _____ Score: _____/75 %: _____ Grade: _____

| Topic Area | Q | Max | Score | % | Priority |
|------------------------|------|-----|-------|---|----------|
| Binary/Hex conversions | 1ab | 3 | | | |
| Hex use | 1d | 2 | | | |
| Image file size | 1ei | 2 | | | |
| Lossy compression | 1eii | 2 | | | |
| ASCII | 2ai | 1 | | | |
| ASCII limitations | 2aii | 2 | | | |
| Sound sampling | 2bi | 2 | | | |
| Sound file size | 2bii | 3 | | | |
| Von Neumann diagram | 3a | 4 | | | |
| Register functions | 3b | 3 | | | |
| FDE trace table | 3d | 2 | | | |
| Memory hierarchy | 4a | 2 | | | |
| Cache | 4b | 2 | | | |

| | | | | | |
|-----------------------|------|---|--|--|--|
| Storage comparison | 4c | 6 | | | |
| Internet vs WWW | 5a | 2 | | | |
| DNS process | 5b | 4 | | | |
| Router vs switch | 5c | 2 | | | |
| IPv6 advantages | 5d | 2 | | | |
| Malware types | 6a | 3 | | | |
| Protection measures | 6b | 6 | | | |
| Sensors/ actuators | 6ci | 2 | | | |
| Automated systems | 6cii | 1 | | | |
| OS functions | 7a | 6 | | | |
| Interrupts | 7b | 2 | | | |
| Packet switching | 8a | 4 | | | |
| Parity | 8bi | 1 | | | |
| Parity explanation | 8bii | 2 | | | |
| Encryption comparison | 8c | 4 | | | |

TOP 5 REVISION PRIORITIES:

102. _____ Action: _____

103. _____ Action: _____

104. _____ Action: _____

105. _____ Action: _____

106. _____ Action: _____

TARGET FOR NEXT MOCK: ____/75 (____%)

CLASS-LEVEL ANALYSIS TEMPLATE (TEACHER USE)

Mock Exam 2 — Paper 1 Date: _____ Class: _____

| Metric | Value |
|-------------------|-------|
| Students sitting | |
| Class average /75 | |
| Class average % | |
| Highest score | |
| Lowest score | |
| A*-A count | |

| | |
|-----------|--|
| B-C count | |
| D-E count | |
| F-G count | |

TOPIC AVERAGES (%):

| Topic | Class Avg % |
|-----------------------|-------------|
| Data Representation | % |
| Text & Sound | % |
| CPU & FDE Cycle | % |
| Memory & Storage | % |
| Networks & Internet | % |
| Security & Tech | % |
| OS & Software | % |
| Transmission & Crypto | % |

TOP 3 CLASS PRIORITIES:

107. _____ Class avg: ____% Action: _____
108. _____ Class avg: ____% Action: _____
109. _____ Class avg: ____% Action: _____

STUDENTS REQUIRING INTERVENTION:

| Student | Score | Grade | Actions |
|---------|-------|-------|---------|
|---------|-------|-------|---------|

COMPARISON WITH MOCK 1:

| | Mock 1 | Mock 2 | Change |
|-----------|--------|--------|--------|
| Class Avg | | | |

NOTES:

END OF DOCUMENT

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